ENHANCING ASTRONOMY LEARNING FOR GRADE 8 STUDENTS IN SRI LANKA THROUGH A STEM-BASED AUGMENTED REALITY MOBILE APPLICATION

Project ID: 25-26J-418

Project Proposal Report

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BSc (Hons) Degree in Information Technology Specializing in Interactive Media

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Declaration

I declare that this is my own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or institute of higher learning. To the best of my knowledge and belief, it does not contain any material previously published or written by another person except where due acknowledgement is made in the text.

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Abstract

Teaching astronomy to middle school students often relies on static textbook diagrams and verbal explanations that fail to convey the dynamic, spatial nature of celestial phenomena. This project proposes an **Augmented Reality (AR)–driven mobile learning application** designed for Grade 8 students in Sri Lanka, combining interactive 3D simulations, gamified quizzes, and teacher monitoring features. The system addresses the limitations of traditional methods by enabling students to visualize and manipulate complex astronomical processes such as **planetary motion, lunar phases, eclipses, and the Sun–Earth–Moon system** in real time through immersive AR content.

The application consists of three integrated components: (1) a **mobile app interface** developed with Android Studio and ARCore that provides students with interactive modules and a teacher dashboard; (2) an **AR content engine** built with Unity and Vuforia to deliver high-quality 3D celestial models and simulations; and (3) a **database and content repository** for storing quizzes, progress records, and teacher feedback. Offline functionality ensures accessibility for schools with limited connectivity, while compatibility with budget Android devices makes the solution scalable and inclusive.

The research methodology involves **requirement analysis, AR content design, quiz integration, system development, and iterative testing in classroom-like settings**. Data will be collected through **student pre- and post-tests (achievement and motivation surveys)** and **teacher surveys and interviews**. Quantitative analysis (paired and independent t-tests) will measure learning gains, while qualitative thematic analysis will capture teachers' perceptions of usability and pedagogical value.

Expected outcomes include improved conceptual understanding, higher student engagement, and positive teacher feedback regarding curriculum integration. By merging **STEM education principles** with emerging AR technologies, this solution aims to **revolutionize astronomy education in Sri Lankan classrooms**, offering a pragmatic, engaging, and scalable approach to science learning

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1. Introduction

Astronomy is one of the oldest sciences, offering students a window into the vastness of the universe while fostering critical thinking, spatial reasoning, and scientific curiosity. However, traditional methods of teaching astronomy in schools often rely on textbook diagrams, static images, and verbal explanations. While useful, these approaches are limited in their ability to convey the dynamic and three-dimensional nature of celestial phenomena. As a result, students frequently struggle with misconceptions regarding planetary motion, lunar phases, eclipses, and the Sun–Earth–Moon system.

Emerging technologies such as Augmented Reality (AR) present powerful opportunities to transform the way astronomy is taught. AR enables learners to interact with 3D models of celestial bodies, visualize abstract processes in real time, and engage in hands-on exploration that would be impossible in a standard classroom. By combining visualization with interactivity, AR bridges the gap between abstract theory and tangible understanding, making learning both engaging and memorable.

This project envisions the design and development of an AR-driven mobile application specifically tailored to the Sri Lankan Grade 8 astronomy curriculum. The solution has two core functions:

- **Interactive AR visualizations** of astronomical concepts such as planetary orbits, eclipses, and lunar phases, allowing students to manipulate models for deeper comprehension.
- **Gamified learning features**, including quizzes with instant feedback and a teacher dashboard for tracking progress and gathering feedback.

Through the integration of these components, the proposed system aims to create an accessible, immersive, and pedagogically aligned learning experience. Offline functionality ensures inclusivity for schools with limited internet access, while compatibility with budget-friendly Android devices makes the solution scalable across diverse educational contexts. Ultimately, this approach seeks to modernize astronomy education in Sri Lanka by merging STEM pedagogy with cutting-edge AR technology, enhancing both conceptual understanding and student engagement.

1.1 Background and Literature Survey

STEM Education and the Role of AR

STEM education is increasingly recognized as essential for preparing students for a technologically advanced world. Beyond subject knowledge, it emphasizes critical thinking, problem-solving, and collaboration (Bybee, 2010). To foster these skills, hands-on and inquiry-based methods are encouraged over rote memorization.

Digital technologies have become central to STEM pedagogy, with Augmented Reality (AR) offering particular promise. AR overlays digital models and simulations onto real-world environments, enabling learners to visualize abstract processes interactively (Bacca et al., 2014). Research shows that AR-supported lessons improve understanding, motivation, and retention (Ibáñez & Delgado-Kloos, 2018). This is especially valuable in subjects like astronomy, chemistry, and physics, where spatial and dynamic phenomena are otherwise difficult to grasp.

Challenges in Astronomy Education

Astronomy poses unique learning challenges due to its reliance on vast scales and invisible processes. Middle school students often struggle with concepts such as planetary motion, eclipses, and lunar phases, frequently holding misconceptions such as attributing lunar phases to Earth's shadow (Bray & Tangney, 2017).

In Sri Lanka, these difficulties are compounded by resource constraints. Grade 8 astronomy instruction primarily relies on textbooks and lectures, with minimal access to models, telescopes, or digital tools. As a result, many students develop incomplete mental models and lose interest in science. Addressing these issues requires interactive tools that make abstract astronomical concepts more tangible.

Existing AR Astronomy Applications

Globally, applications such as *Star Walk 2*, *Solar System AR*, and *SkySafari* demonstrate how AR can make astronomy engaging through virtual models and simulations. However, most are designed for general audiences and lack curriculum alignment, reducing their classroom utility.

Research shows that AR astronomy tools are largely concentrated in Western or technologically advanced settings (Omurtak & Zeybek, 2022; Keçeci et al., 2021). Few studies address their adaptation to developing countries, where factors such as language, curriculum requirements, and cultural context significantly shape adoption. This underscores the need for localized AR solutions for Sri Lankan classrooms.

Teacher Perspectives

Teachers play a decisive role in the adoption of new technologies. Their involvement ensures alignment with curriculum goals and enhances sustainability (Jong et al., 2013). However, in Sri Lanka, teacher input is rarely considered in the design of digital tools, and professional support is limited. This disconnect often results in low or inconsistent classroom usage, even when AR resources are available. For effective implementation, teacher perspectives must be integrated from the outset.

Conclusion

The literature highlights AR's potential to improve STEM education by enhancing conceptual understanding and engagement. In astronomy, AR helps address misconceptions and makes invisible processes accessible. However, most tools remain generic, not curriculum-specific, and overlook teacher perspectives. In Sri Lanka, there is a clear gap in localized, curriculum-aligned AR applications for Grade 8 astronomy. This study addresses this gap by developing and evaluating an AR mobile app tailored to the national curriculum, incorporating teacher feedback to ensure pedagogical relevance and classroom feasibility.

1.2 Research gap

Over the past decade, Augmented Reality (AR) has been increasingly explored in educational contexts, particularly in STEM subjects. While prior studies demonstrate AR's potential to enhance visualization, motivation, and engagement, several limitations remain that restrict its effectiveness in classroom integration.

Firstly, many AR-based learning solutions are **siloed in scope**, focusing either on visualization of 3D models, quiz-based interactivity, or basic engagement tracking, but rarely offering an **integrated system** that combines interactive simulations, gamified learning, teacher dashboards, and offline accessibility. This lack of holistic design limits the adaptability of such solutions in real-world schools, especially in developing regions with budget and connectivity constraints.

Secondly, much of the existing research prioritizes **qualitative outcomes** such as student perceptions and engagement (e.g., surveys, interviews) while neglecting **rigorous quantitative validation** of learning gains. Studies often report descriptive improvements without applying statistical methods to test significance. This creates a gap in **empirical evidence** about the actual impact of AR on student achievement.

Finally, to date, very few AR-in-education projects have employed **formal statistical methods** such as *paired t-tests* or *independent t-tests* in their evaluation design. Without such controlled analysis, it remains unclear whether AR interventions measurably outperform traditional teaching methods in improving conceptual understanding.

This research addresses these gaps by (1) developing a curriculum-aligned, all-in-one AR mobile application with offline functionality, (2) integrating teacher dashboards for progress monitoring, and (3) applying a robust experimental design with paired and independent t-tests to provide statistically validated evidence of AR's effectiveness in improving astronomy learning outcomes among Grade 8 students.

Feature	**Star Walk 2**	**Solar System AR**	**SkySafari**	**Omurtak & Zeybek, 2022**	**Keçeci et al., 2021**	**Proposed Solution**
AR-based 3D Visualization	Yes	Yes	Yes	Yes	Yes	Yes
Curriculum Alignment	No	No	No	Partial	Partial	Yes
Teacher Dashboard / Monitoring	No	No	No	No	No	Yes
Offline Functionality	No	No	No	No	No	Yes
Gamified Learning Modules	No	No	No	No	No	Yes
Integrated All-in-One System	No	No	No	No	No	Yes
Statistical Validation of Learning Gains (Paired t-test)	No	No	No	No	No	Yes

Table 1 – Comparison of Gaps

1.3 Research Problem

Despite growing interest in digital STEM tools, astronomy education for middle school students in Sri Lanka still relies heavily on textbooks, diagrams, and lectures, which inadequately convey dynamic, three-dimensional celestial phenomena and often lead to persistent misconceptions. Although AR has shown global potential for enhancing engagement and understanding, most existing applications target general audiences, lack alignment with national curricula, ignore teacher perspectives, and require high-end devices or constant internet, limiting accessibility in typical Sri Lankan classrooms.

The challenge, therefore, is to design and develop a **curriculum-aligned AR mobile application for Grade 8 astronomy** that:

- Enables students to visualize and interact with complex astronomical processes in an immersive, real-time 3D environment.
- Integrates assessment modules such as interactive quizzes to measure learning progress.
- Incorporates teacher dashboards and feedback mechanisms to ensure pedagogical relevance and classroom feasibility.
- Functions offline and on budget-friendly Android devices, ensuring accessibility and scalability.
- Employs rigorous experimental validation, such as pre-test/post-test and paired t-tests, to quantify learning gains.

This research aims to bridge the gap between traditional teaching limitations and modern AR-enabled learning by creating an interactive, inclusive, and scientifically validated tool that enhances both student understanding and teacher confidence in astronomy education.

2. Objectives

2.1 Main Objective

The primary goal of this study is to design, develop, and evaluate a STEM-based Augmented Reality (AR) mobile application that enhances astronomy learning for Grade 8 students in Sri Lanka. The application targets abstract concepts such as planetary motion, lunar phases, and eclipses, providing an interactive and immersive experience to improve comprehension, engagement, and knowledge retention. Additionally, the study aims to assess the practical feasibility of classroom integration from the teachers' perspective.

2.2 Sub Objectives

- Identify teacher and student perceptions of current astronomy teaching methods

  Gather insights on challenges, misconceptions, and attitudes toward digital learning to inform app design.

- Design an AR application aligned with the Grade 8 curriculum

  Develop interactive 3D models, simulations, and activities that directly support national learning outcomes and classroom instruction.

- Evaluate application effectiveness using pre-test/post-test and paired t-tests

  Measure student learning gains and provide quantitative evidence of the AR app's educational impact.

- Analyze teacher feedback on usability and classroom integration

  Collect surveys and interviews to assess the app's practicality, user-friendliness, and alignment with lesson plans.

- Compare AR-based learning with traditional textbook instruction

  Examine differences in engagement, understanding, and knowledge retention to support recommendations for technology-enhanced STEM learning.

3. Methodology

System Overview

The proposed system is designed as a comprehensive Augmented Reality (AR) learning platform for Grade 8 astronomy students, integrating interactive content, performance tracking, and teacher support. The system is composed of three interconnected components—front-end mobile application, AR content engine, and database/content storage—that collectively provide an immersive, curriculum-aligned learning experience. Figure X illustrates the overall system architecture, showing the flow of student interactions, AR simulations, and teacher monitoring.

Front-End Mobile Application (Android Studio + ARCore)

The mobile application acts as the primary interface for both students and teachers. Students can access interactive lesson modules, 3D AR simulations, and embedded quizzes that reinforce learning. Through the app, students can manipulate virtual representations of planets, the Sun–Earth–Moon system, lunar phases, and eclipses, making abstract concepts tangible and interactive. Teachers access a dashboard to monitor student engagement, quiz performance, and progression, enabling data-driven instructional decisions. The app is designed to be user-friendly and requires minimal training.

AR Content Engine (Unity + Vuforia)

The AR engine delivers high-quality 3D visualizations and interactive simulations. Unity allows the creation of detailed celestial models, while Vuforia ensures accurate AR tracking in real-world settings. Students can rotate, scale, and explore virtual objects to comprehend spatial relationships and dynamic astronomical processes, such as orbital motion and eclipses. This component bridges the gap between abstract textbook concepts and experiential learning, enhancing understanding and retention.

Database and Content Storage

The database manages all student data, quiz results, and teacher feedback securely. It also stores AR content and lesson modules, enabling educators to update materials and quizzes to remain aligned with the curriculum. Tracking student performance allows for data-driven evaluation of learning outcomes and the effectiveness of the AR application.

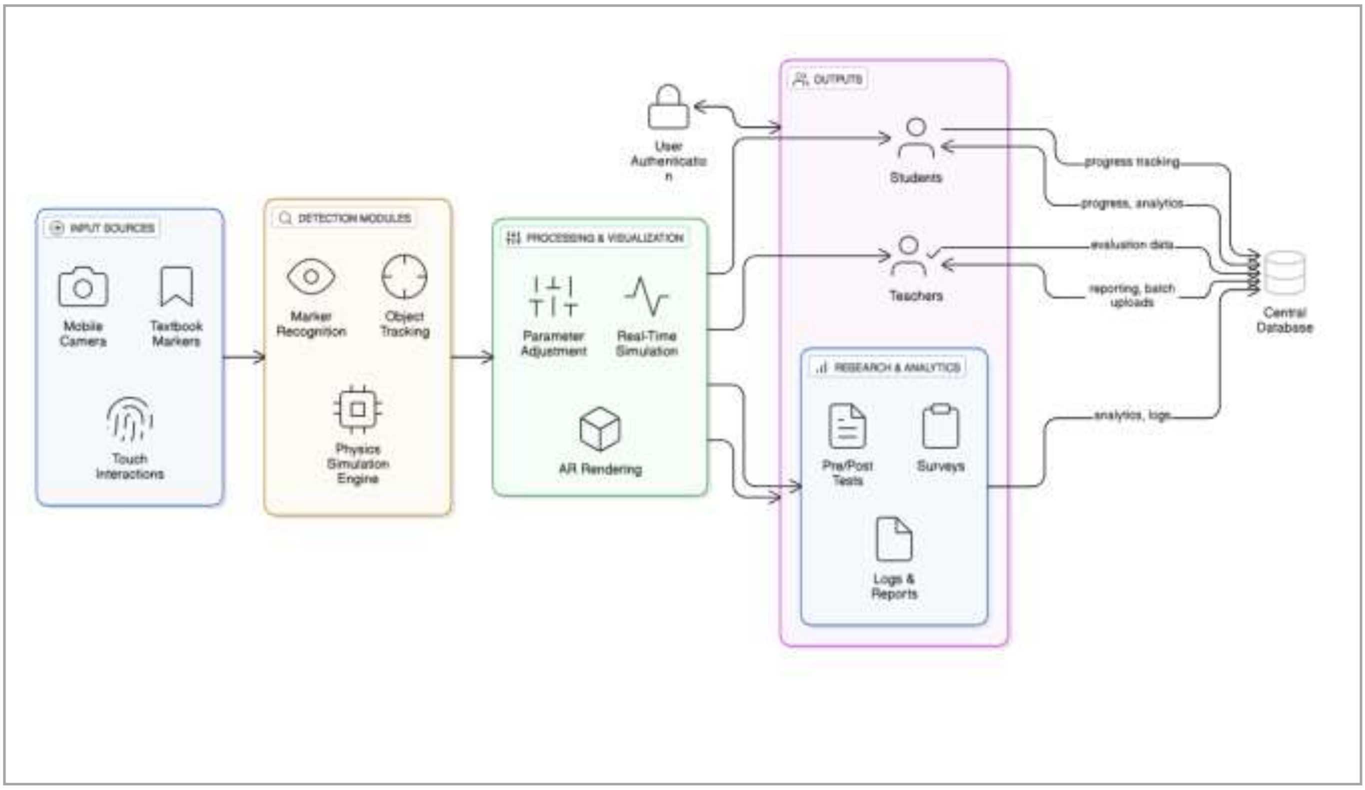

Figure 1- system diagram

Participants

The study involves two primary groups

- **Students:** 30–50 Grade 8 English-medium students from a selected school will participate, engaging with AR modules while their learning gains are assessed.
- **Teachers:** 2–3 geography or science teachers teaching Grade 8 astronomy will provide feedback on usability, curriculum alignment, and classroom integration.

Data Collection

A mixed-methods approach will be employed:

- **Student Data:** Pre-tests and post-tests assessing achievement and motivation will be conducted to evaluate the educational impact of the AR application.
- **Teacher Data:** Structured surveys and semi-structured interviews will gather qualitative insights regarding usability, classroom feasibility, and pedagogical alignment.

Data Analysis

- **Students:** Paired t-tests will analyze pre-test and post-test scores within the AR group to determine individual learning gains, while independent t-tests will compare results between AR and traditional instruction groups.
- **Teachers:** Thematic analysis of interview data will identify key themes related to usability, classroom integration, and perceived educational benefits of the AR application.

Research Phases

The project will proceed through five systematic phases:

1. **Requirement Analysis:** Gather curriculum requirements, teacher/student perceptions, and design specifications for AR content.

2. **AR Content Development:** Create 3D models and interactive simulations for celestial phenomena.
3. **App and Dashboard Development:** Integrate interactive modules, quizzes, and teacher monitoring features.
4. **Pilot Testing:** Conduct usability testing in classroom-like environments, collecting student performance and teacher feedback.
5. **Evaluation and Refinement:** Analyze data, refine AR content and interface, and validate learning outcomes.

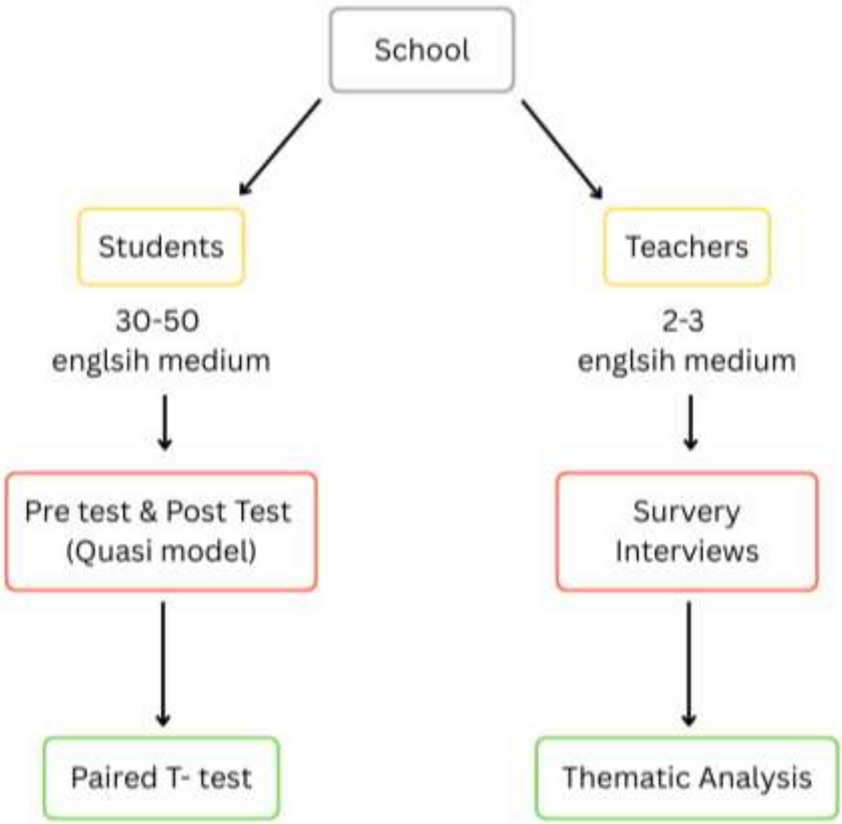

Figure 2 – Analyzing methods

3.1 Project Requirements

Functional Requirements

AR Visualization of Astronomy Concepts

The system must enable students to interact with Augmented Reality (AR) models of key Grade 8 astronomy topics, including planets, Moon phases, the Sun–Earth–Moon system, and eclipses. These models should be manipulatable (rotate, zoom, scale) to allow students to explore spatial relationships and dynamic processes in a hands-on manner.

Interactive Quizzes with Immediate Feedback

The application will integrate quizzes within each lesson module. After attempting questions, students will receive instant feedback highlighting correct answers, explanations, and areas for improvement. This ensures active learning and self-assessment.

Teacher Dashboard for Monitoring

A lightweight teacher dashboard must be included to track student progress. Teachers will be able to view quiz scores, monitor lesson completion rates, and identify areas where students struggle. This provides data-driven insights to support classroom teaching.

Offline Functionality

Once the application and AR content are initially downloaded, it should function offline without requiring continuous internet access. This ensures accessibility in schools with limited or unstable connectivity.

Non-Functional Requirements

Cross-Device Compatibility

The application must be optimized to run smoothly on budget-friendly Android devices commonly used in Sri Lankan schools. This ensures inclusivity and prevents technological barriers to adoption.

User-Friendly Interface

The system should feature a simple and intuitive interface that requires minimal training for both students and teachers. Navigation must be clear, with age-appropriate icons, instructions, and interactive elements.

Smooth 3D Rendering and AR Interaction

The AR simulations and 3D models must be optimized for performance, providing smooth rendering and responsive interaction. Lag, jitter, or crashes should be minimized to ensure an engaging and frustration-free learning experience.

4. References

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5. Appendices

- Sample teacher survey questionnaire.
- Sample student pre-test and post-test questions.
- Mockups of AR app interface screens.
- 3D models.

6. Work Breakdown Structure

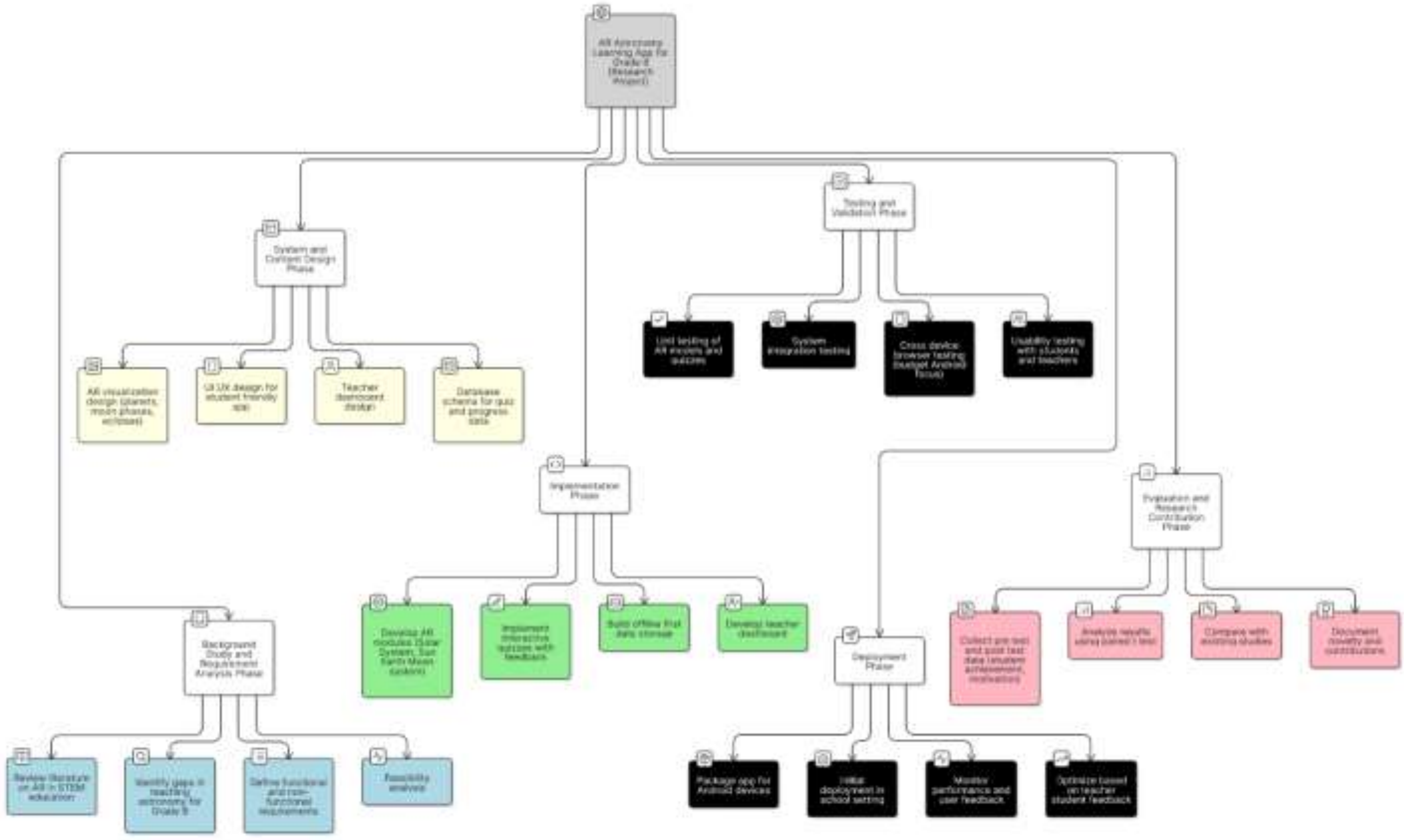

Figure 3 – Work Breakdown Structure